

(* Dynamical Systems, Fall 2025 - HW03 - Question 3 - Marcello Ghini Bettiol *)

(* Goal:

find an approximation to the unstable manifold of the system provided by using

$$-2h_u + x^2 + h_u^2 = h_u' \cdot (2x + h_u^2)$$

and a power series approximation for h_u up to $O(x^6)$. We write

$$h_u(x) = a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots,$$

where $a_0 = a_1 = 0$ because $h_u(0) = h_u'(0) = 0$.

*)

In[819]:=

```
maxorder = 6;
```

```
h[x_] := a2 * x^2 + a3 * x^3 + a4 * x^4 + a5 * x^5 + a6 * x^6;
```

(* Equation for h_u *)

```
eq = -2 * h[x] + x^2 + h[x]^2 == D[h[x], x] * (2 * x + h[x]^2);
```

(* Take the series expansion of the equation for h_u *)

```
serieeq = Series[eq, {x, 0, maxorder}] // Normal // Expand;
```

(* Match coefficients *)

```
coefs = CoefficientList[serieeq /. Equal -> Subtract, x];
```

```
coefs = Thread[coefs == 0];
```

(* Solve the system for the coefficients *)

```
sol = Solve[coefs, {a2, a3, a4, a5, a6}]
```

```
huapprox[x_] = h[x] /. sol[[1]];
```

```
Simplify[huapprox[x]];
```

```
Print["Local unstable manifold (x, h_u(x)), h_u(x)=", huapprox[x]]
```

(* As in Example 5.12 of Meiss,

the local stable manifold is given by $(h_s(y), y)$, where

$$h_s(y) = -y^2/6 * (1 + y/4 + y^2/40).$$

*)

```
hsapprox[y_] := -y^2 / 6 * (1 + y / 4 + y^2 / 40)
```

(* Plot the local unstable manifold over

the phase portrait of the nonlinear system *)

```
f[x_, y_] := 2 * x + y^2;
```

```
g[x_, y_] := -2 * y + x^2 + y^2;
```

```

umnfd = Table[{x, huapprox[x]}, {x, -1, 1, 0.001}];
smnfd = Table[{hsapprox[y], y}, {y, -1, 1, 0.001}];

pp = StreamPlot[
  {f[x, y], g[x, y]}, {x, -1, 1}, {y, -1, 1},
  StreamColorFunction -> None,
  StreamStyle -> Gray,
  StreamPoints -> Fine,
  FrameLabel -> {"x", "y"},
  PlotRange -> {{-1.1, 1.1}, {-1.1, 1.1}},
  PlotLabel ->
    Style["Phase Portrait with local stable and unstable manifolds", Bold, 16],
  ImageSize -> Large,
  Epilog -> {{Thick, Blue, Line[smnfd]}, {Thick, Red, Line[umnfd]}}
];

fig = Legended[pp, Placed[
  LineLegend[{Blue, Red}, {"local stable manifold", "local unstable manifold"},
  LegendMarkerSize -> 20, LabelStyle -> {FontSize -> 14}], Right]]

Export[FileNameJoin[{NotebookDirectory[], "phase_portrait.png"}], fig];

```

Out[825]=

$$\left\{ \left\{ a_2 \rightarrow \frac{1}{6}, a_3 \rightarrow 0, a_4 \rightarrow \frac{1}{360}, a_5 \rightarrow -\frac{1}{1296}, a_6 \rightarrow \frac{1}{15120} \right\} \right\}$$

$$\text{Local unstable manifold } (x, h_u(x)), \quad h_u(x) = \frac{x^2}{6} + \frac{x^4}{360} - \frac{x^5}{1296} + \frac{x^6}{15120}$$

Out[835]=

